

- 1 (a) work done in bringing/moving unit massM1
 from infinity to the point..... A1 [2]
 (use of 1 kg in the definition – max 1/2)
- (b) potential at infinity defined as being zero..... B1
 forces are always attractive..... B1
 so work got out in moving to point..... B1 [3]
 (max potential is at infinity – allow 1/3)
- (c) (i) change = $6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (\{6.4 \times 10^6\}^{-1} - \{1.94 \times 10^7\}^{-1})$ C2
 change = $4.19 \times 10^7 \text{ J kg}^{-1}$ (ignore sign)A1
- (ii) $\frac{1}{2}mv^2 = m\Delta\phi$ C1
 $v^2 = 2 \times 4.19 \times 10^7 = 8.38 \times 10^7$
 $v = 9150 \text{ m s}^{-1}$ A1 [5]
- (d) acceleration is not constant..... B1 [1]

2 (a) 